

Supporting Adult Guidance

Using Index Laws · Protecting pattern-finding, agency and a calm return to prerequisite ideas

Connection before curriculum. Always.

The adult protects thinking and regulation. Invite the learner to see repeated factors before supplying a rule, and use a Reconnection Route only when a specific prerequisite is blocking entry.

Lesson purpose at a glance

The learner selects and applies the multiplication, division and power-of-a-power index laws, connects them to repeated multiplication and encounters the zero-index pattern through division.

What the learner is developing

- accurate use of base and exponent language;
- structural attention to matching bases;
- selection of a law from the operation;
- reasoning from repeated factors;
- confidence to diagnose common misconceptions.

Key mathematical ideas

- $a^m \times a^n = a^{m+n}$: two groups of matching factors join;
- $a^m \div a^n = a^{m-n}$: matching factors cancel;
- $(a^m)^n = a^{mn}$: a group of m factors is repeated n times;
- $a^0 = 1$ for $a \neq 0$: a power divided by itself gives both a^0 and 1.

Preparation and Reconnection Routes

Essential preparation

- Confirm that Guidance, Reconnection Routes, Scratchpad and Practice Companion controls are visible.
- Have paper or another recording option available.
- Check that the learner knows they may pause or continue later.

Reconnection is attached to need

Reconnection is not a catch-up block. Use the in-page button only when multiplication, powers as repeated multiplication, base/exponent language or an operation symbol is preventing entry to the current task.

1. Name the blocking idea neutrally.
2. Invite the shortest relevant route.
3. Allow an answer, sketch or hint.
4. Record only the minimum return evidence.
5. Use Return to Using Index Laws and resume the original task.

Useful language

“Would a short powers route help here?” “You can return to the Pattern Machine as soon as that one connection feels secure enough.”

Supporting the six Cornerstones

Connection

Ask the learner to expand $2^3 \times 2^4$ and count the factors before naming the law.

Movement

Invite a change to one exponent and ask how the simplified exponent moves.

Reflection

Ask which operation is present and why that selects the law.

Creativity

Invite three expressions with one shared target power.

Rest

Use the calm retrieval route: same base → operation → exponent rule.

Nutrition

Name intellectual nourishment honestly: structure, reasoning, efficiency and precision.

Invitational rhythm

Connect 8 · Move 18 · Practise 10 · Reflect/Create 5 · Rest/Nourish 2 · Evidence 2. These are orientation points, not deadlines.

Common difficulties and supportive responses

The learner changes the base.

Return to repeated factors and identify which number is repeated.

The learner multiplies exponents when multiplying powers.

Expand a small example and count all factors in the joined groups.

The learner adds exponents for a power of a power.

Represent several identical groups and count factors per group $\times$ number of groups.

The bases do not match.

Pause before using a law. These three laws require a common base in their current form.

The zero exponent feels impossible.

Compare $a^6 \div a^6 = 1$ with $a^{6-6} = a^0$.

The learner appears overloaded.

Reduce the visible task, offer another recording mode, or pause without converting the pause into failure.

Practice Companion support pathway

Universal worked-solution release rule

For every question independently: Socratic prompt used + Hint used + two non-empty answers entered and checked = Worked solution released.

6. Allow time to think, sketch or begin independently.
7. Invite the Socratic prompt if useful.
8. Let the learner enter and check a first answer.
9. Invite the Hint if useful.
10. Let the learner enter and check a second answer.
11. Only then may the Worked solution unlock for that question.

Do not provide the complete worked solution verbally while the button is locked. You may clarify vocabulary, read the question aloud, support regulation or invite a Reconnection Route without supplying the method.

QA observation

- Prompt, hint and attempts reset independently for each question.
- An empty answer does not count as an attempt.
- Navigation retains the correct state for each question.
- The zero-index response accepts 0 as the resulting exponent.

Access, evidence and a valid finish

Access and regulation

- Use age-neutral language and avoid deficit framing.
- Allow spoken, typed, drawn or annotated responses.
- Keep the Scratchpad optional and accept paper or another tool.
- Do not use time-pressure language.
- Treat incomplete work as information for the next connection.

Evidence to notice

- Does the learner check that bases match?
- Can they name the operation before choosing a law?
- Can they calculate the resulting exponent accurately?
- Can they preserve the base?
- Can they explain one law using repeated factors?

A valid finishing point

The learner may finish after one accurate law selection, one clear repeated-factor explanation or one identified Reconnection Route and return point. The next lesson decision follows evidence and regulation, not a fixed completion demand.

Observation notes

Secure connection noticed: _____

Reconnection Route used and return evidence:

Next mathematical connection:

Access or regulation support that helped:
